UE24CS352A- Machine Learning
Mini-Project Assignment

Assigned Problem: Predicting Hit Songs from Audio Features

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1. Problem Statement

Inspired by the CS229 project Predicting Hit Songs Using Repeated Chorus (Liu, 2021), our project investigates whether machine learning can predict song popularity using only audio characteristics. We formulate this as a binary classification problem: a song is a hit (1) if its FMA listen count is in the top 25%, and not a hit (0) otherwise.

The aim is to determine whether a short audio excerpt contains enough information to distinguish popular songs from less popular songs, while recognizing that factors such as marketing, artist reputation and social trends also influence success.

2. Dataset Details

We use the Free Music Archive (FMA) dataset, specifically the FMA-small subset. It contains 8,000 tracks from 2,309 artists, with each track represented by a 30-second mid-song excerpt. Each track is described using 518 audio features generated using librosa. These include MFCCs, chroma STFT/CQT/CENS, tonnetz, spectral centroid, spectral bandwidth, spectral contrast, spectral rolloff, RMS energy, and zero-crossing rate. Each feature is summarised using seven statistics: mean, standard deviation, skewness, kurtosis, median, minimum, and maximum.

The target label is created from FMA listen counts. Tracks in the top 25% of listen counts are labelled hits, resulting in 2,000 hits and 6,000 non-hits. This produces an imbalanced dataset, so accuracy alone is not a reliable measure of performance.

There are important differences from the original CS229 project. The original study used 554 Billboard songs and extracted repeated choruses, whereas our project uses the larger FMA dataset and 30-second mid-song excerpts. We also define a hit using FMA listen counts rather than Billboard chart status.

3. Approach Used to Solve the Problem

The first step is data preparation, where the FMA metadata and feature files are combined and the hit/non-hit labels are generated. We then use an 80/20 artist-wise train-test

split. This ensures that no artist appears in both training and testing, preventing the model from simply learning artist-specific popularity patterns and providing a more realistic measure of generalisation.

For most models, the preprocessing pipeline is standardisation → PCA → classifier. StandardScaler normalises the feature values, while PCA retains 95% of the variance, reducing the original 518 features to 205 principal components. Scaling and PCA are performed within the cross-validation pipeline to avoid data leakage. Tree-based models use the original 518 features.

We compare several supervised learning algorithms: Logistic Regression with Elastic Net, LDA with shrinkage, Linear SVM, RBF SVM, Polynomial SVM, Random Forest, Gradient Boosting, and a Neural Network (MLP). We also use two baselines: always predicting “not hit” and random guessing.

Hyperparameters are selected using GridSearchCV with 5-fold GroupKFold cross-validation, with artist IDs used as groups. Models are optimised using ROC-AUC, and balanced class weights are used where applicable to reduce the effect of the 75:25 class imbalance.

4. Brief Overview of Our Implementation

The project is implemented in Python using Pandas, NumPy, Scikit-learn, Librosa and Joblib. prepare_data.py loads the FMA metadata and features, creates the hit labels, and produces the processed dataset. train.py performs the artist-wise split, preprocessing, model training, hyperparameter tuning, and evaluation. The project also includes scripts for generating plots and making predictions from new audio files.

Performance is evaluated using accuracy, F1-score, and ROC-AUC, with greater emphasis on F1 and ROC-AUC because of class imbalance. The RBF SVM achieved the highest cross-validation ROC-AUC of 0.711 and was selected based on cross-validation performance. On the held-out test set, Gradient Boosting achieved the highest ROC-AUC of 0.695, while Random Forest achieved the highest accuracy of 0.792. However, Random Forest had low recall for actual hits, showing why accuracy alone can be misleading.

Feature-family analysis showed that MFCC features were the most influential, producing the largest drop in test ROC-AUC when shuffled (0.085), followed by chroma STFT (0.039) and spectral contrast (0.030). This suggests that timbre-related characteristics contain useful information for predicting popularity.

5. Conclusion

The results provide partial support for using audio characteristics to predict song popularity. The models achieved test ROC-AUC values of approximately 0.64–0.70, clearly above the baseline of about 0.50, showing that audio contains useful predictive information.

However, the moderate performance shows that audio alone cannot reliably predict a hit. Artist popularity, lyrics, marketing, genre and social trends also influence success. Future work could combine audio with these additional factors to build a more comprehensive song-popularity prediction system.